

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 36)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Table 1 shows some radioactive isotopes, their half-lives and the type of radiation they emit.

Table 1

Isotope	Half-life	Main radiation	Source
Iodine-131	8 days	Beta / gamma	Fission in reactors
Strontium-90	29 years	Beta	Fission in reactors
Caesium-137	30 years	Beta / gamma	Fission in reactors
Plutonium-239	24 400 years	Alpha	Neutron bombardment of U-238
Uranium-238	4.5×10^9 years	Alpha	Occurs naturally in rocks

0 1

1

Define the term **half-life**.

[1 mark]

0 1

2

A sample of iodine-131 has an activity of 6400 Bq. Calculate its activity after 32 days.

[2 marks]

Answer

Bq

0 1

3

Use Table 1 to explain why isotopes with half-lives of several decades, such as caesium-137, cause the greatest concern after a nuclear accident.

[3 marks]

0 1 . 4

Explain why plutonium-239 is relatively safe outside the body but very dangerous if inhaled or ingested.

[3 marks]

0 1 . 5

Explain what is meant by an **activation product**.

[1 mark]

10

0 2

A nuclear fuel reprocessing plant discharges small amounts of radioactive material in liquid effluent to a coastal sea.

0 2 . 1

Explain why it cannot be assumed that the discharge will be diluted to insignificant levels.

[2 marks]

0 2 . 2

Give **four** factors that would be included in a Critical Pathway Analysis of this discharge.

[4 marks]

1

2

3

4

0 2 . 3

Explain how the results of the analysis reduce the cost of environmental monitoring.

[2 marks]

0 2 . 4

Explain why Critical Group Monitoring measures exposure rather than damage to health.

[2 marks]

10

03 . 1

Describe how DNA databases are used to control the illegal trade in tropical timber and ivory.

[3 marks]

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03 . 2

Explain the difference between radio / GPS tracking and acoustic tracking and state when each is used.

[2 marks]

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03 . 3

Give **three** types of carrier system used to move sensors to study areas and one advantage of drones for ecological monitoring.

[3 marks]

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03 . 4

Explain how motion-sensitive cameras and CCTV reduce disturbance to wildlife compared with direct observation.

[2 marks]

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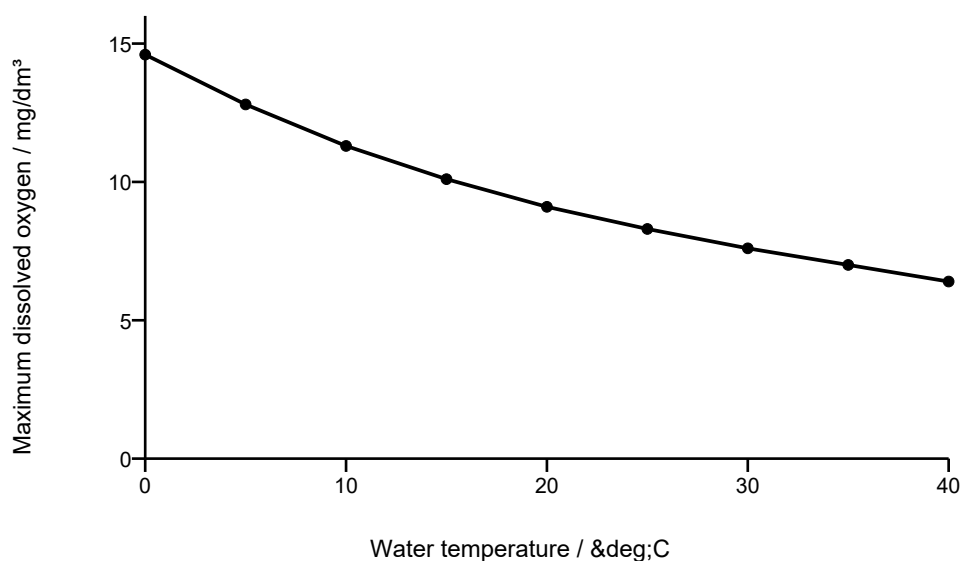
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10

0 4

Figure 1 shows the maximum concentration of dissolved oxygen in fresh water at different temperatures. A power station discharges warm cooling water into a river. The lowest dissolved oxygen concentration that trout can tolerate is 6.5 mg dm^{-3} .

Figure 1



0 4

1

Describe the relationship shown in Figure 1.

[1 mark]

0 4

2

Use Figure 1 to determine the highest water temperature at which trout could survive if the water were fully saturated with oxygen.

[1 mark]

Answer

&deg;C

0 4 . 3

In summer the river upstream of the power station is at 20 °C and the effluent raises the temperature by 7 °C. Use Figure 1 to calculate the reduction in maximum dissolved oxygen concentration caused by the effluent.

[2 marks]

Answer

mg/dm³

0 4 . 4

Explain why the actual dissolved oxygen concentration downstream may be much lower than the values in Figure 1.

[2 marks]

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0 4 . 5

Describe **three** effects of warmer water on aquatic organisms other than reduced dissolved oxygen.

[3 marks]

1

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2

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3

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0 4 . 6

Describe how a cooling tower reduces the temperature of power station effluent water.

[1 mark]

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10

0 5

A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

0 5**1**

Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

[2 marks]

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0 5**2**

Explain the conditions that would need to be met for the site to be used for housing or farmland.

[2 marks]

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0 5**3**

Explain why moving the wildlife from a site before mining begins is rarely completely successful.

[1 mark]

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5**0 6****1**

Complete Table 1 to explain how each method reduces soil erosion.

[5 marks]**Table 1**

Method	How it reduces erosion	
Terracing		
Tied ridging		
Windbreaks / hedgerows		
Mulching		
Strip cropping		

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0	6	.	2
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Explain why long-term crops such as tea, coffee or orchards reduce erosion compared with annual arable crops.

[2 marks]

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0	6	.	3
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Describe **three** ways livestock can be managed to reduce soil erosion.

[3 marks]

10

07

Figure 1 lists the stages in a water treatment works supplying drinking water from a lowland river.

Figure 1

Stage	Process
1	Screens
2	Sedimentation
3	Aeration
4	Flocculation and clarification
5	Filtration through sand and gravel
6	Activated carbon
7	Sterilisation
8	pH adjustment

07**1**

Explain why clay particles do not settle out in the sedimentation stage, and describe how they are removed in stage 4.

[3 marks]

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07**2**

Explain the purpose of aeration (stage 3).

[2 marks]

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07**3**

Explain why ozone or UV light may be used for sterilisation instead of chlorine when water comes from peaty upland sources.

[2 marks]

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0 7 . 4 Describe how each of the following contaminants is removed from water. **[3 marks]**

Toxic metal ions (lead, mercury, arsenic)

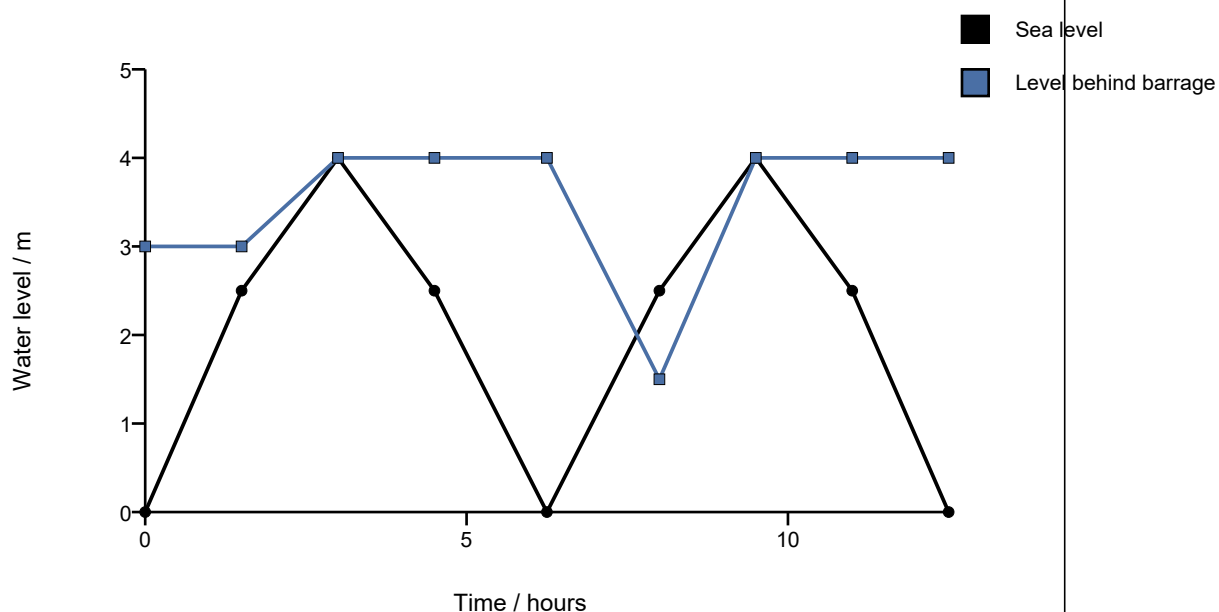
Salt

Odours and pesticides

10

0 8 Figure 1 shows the water level inside and outside a tidal barrage over one 12.5-hour tidal cycle and the periods when electricity is generated.

Figure 1



0 8 . 1 Use Figure 1 to explain when electricity is generated by the barrage. **[2 marks]**

08 . 2

Explain why tidal power is intermittent but predictable.

[2 marks]

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08 . 3

Explain the difference between spring and neap tides and why this affects output.

[2 marks]

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08 . 4

Explain how a tidal lagoon can be used to generate electricity when there is no tidal flow.

[2 marks]

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08 . 5

Give **two** coastal features that make a site suitable for tidal power.

[2 marks]

1

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2

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10

0 9

Two insecticides, X and Y, were tested on a pest insect and on a predatory ground beetle that feeds on the pest. Table 1 shows the results.

Table 1

Insecticide	LD ₅₀ for pest / mg kg ⁻¹	LD ₅₀ for ground beetle / mg kg ⁻¹	Half-life in soil / days	Solubility in water
X	5	6	180	Low
Y	8	150	4	High

0 9**. 1**

Explain what is meant by LD₅₀.

[1 mark]**0 9****. 2**

Using Table 1, explain which insecticide is more likely to cause indirect effects on the pest population through the food chain.

[3 marks]**0 9****. 3**

Suggest **two** environmental problems that could be caused by insecticide Y that are less likely with insecticide X.

[2 marks]**1****2****0 9****. 4**

Explain why the half-life of insecticide X makes it more likely to bioaccumulate.

[2 marks]

0	9	.	5
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Give **two** reasons why a systemic insecticide may be preferred to a contact insecticide.

[2 marks]

10

1	0	.	1
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Evaluate the extent to which new technologies could allow fossil fuels and nuclear fission to continue to supply a large proportion of the world's energy sustainably.

[9 marks]

Extra space

10 . 2

State the approximate percentage of natural gas in a reservoir that can be recovered using its own pressure.

[1 mark]

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Write an essay on one of the following topics.

1	1	.	1
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Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

OR

1 1 . 2

Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

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Handwriting practice area with 25 horizontal dotted lines.

END OF QUESTIONS